NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 17/01/2026
Time: 3 Hrs | GTM – 17 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	A	B	C	B	D	C	A	D	B
11	12	13	14	15	16	17	18	19	20
A	D	C	C	A	A	C	B	A	C
21	22	23	24	25					
6	11	1	79	4					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	D	C	A	C	C	A	D	B	A
36	37	38	39	40	41	42	43	44	45
C	C	D	B	D	B	C	C	A	A
46	47	48	49	50					
75	4	10	12	34					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
A	C	C	A	C	B	A	B	D	C
61	62	63	64	65	66	67	68	69	70
D	C	B	D	B	D	A	A	C	C
71	72	73	74	75					
45	46	2	90	42					

Rankers Academy JEE

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HINTS
PHYSICS

1. No. of fringes = $\frac{length\ of\ segment}{fringe\ width}$

$N_1\lambda_1 = N_2\lambda_2$

$12\times 600 = N_2\times 400$

$N_2 = 18$

2. frequency=speed/wavelength
3. In conductor free electrons are moving randomly in all the possible direction therefore net force acting on it zero
4. velocity of block is one and half times the velocity of the spherical body.

$V_b = 3R\omega$

$K.E_b = \frac{1}{2}mV_b^2$

$K.E_{sphere} = \frac{1}{2}I_{IOR}\omega^2$

$= \frac{1}{2}6mR^2\frac{V_b^2}{9R^2}$

$= \frac{mV_b^2}{3}$

5. $B_{net} \propto \frac{1}{r}$
6. only translational kinetic energy is 1.5 pV.
7. Electric potential due to a charged disc at its centre is $\frac{\sigma R}{2\epsilon_0}$. The potential will be same for nth part of the disc is $\frac{\sigma R}{2\epsilon_0 n}$.
8. $a = -\frac{dv}{dt}$

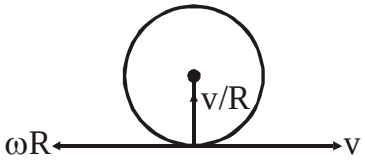
$kv^2 = -\frac{dv}{dt}$, $\int_0^t kdt = \int_{v_0}^{v} -\frac{dv}{v^2}$

$kt + \frac{1}{v_0} = \frac{1}{v}$

$v = \frac{v_0}{1+kv_0t}$

9. As the disc is in combined rotation and translation, each point has a tangential velocity and a linear velocity in the forward direction.

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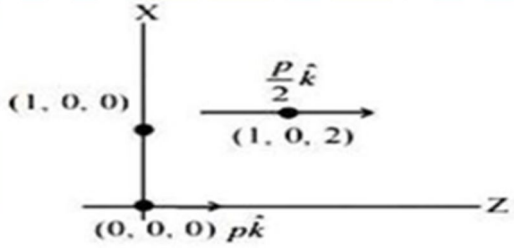


From figure, v_{net} (for lowest point) $= v - R\omega = v - v = 0$ and acceleration $\frac{v^2}{R} + 0 = \frac{v^2}{R}$ (since linear speed of constant)

10. For pk it is equatorial point

$$\therefore \overline{E_1} = \frac{1}{4\pi\varepsilon_0}\frac{P}{1}\left(-\hat{k}\right) \qquad \therefore \overline{E_2} = \frac{1}{4\pi\varepsilon_0}\frac{\frac{P}{2}\times 2\left(\hat{k}\right)}{2^3} = \frac{1}{4\pi\varepsilon_0}\frac{p}{8}\hat{k}$$

For $\frac{P}{2}\hat{k}$ it is axial point $\qquad \therefore \overline{E} = \overline{E_1} + \overline{E_2} = -\frac{7p}{32\pi\varepsilon_0}\hat{k}$



11. Solution : $B_L = \frac{\mu_0 I}{2\pi R}$

If the radius is R/n; the number of turns will be n.

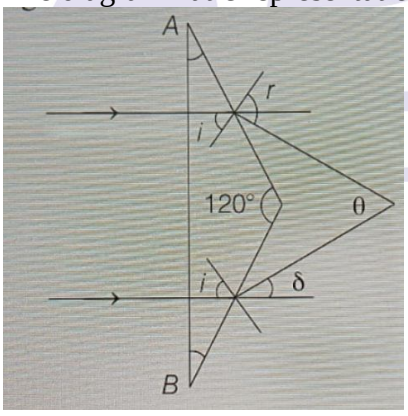
$$B_C = \frac{n\mu_0 I}{2\pi\left(R/n\right)} = n^2\frac{\mu_0 I}{2\pi R}$$

Hence, $\frac{B_L}{B_C} = \frac{1}{n^2}$

12. Conceptual

13. Conceptual

14. The diagrammatic representation of the given problem is shown in figure.



From figure it follows that $\angle i = \angle A = 30^0$

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From Snell's law, $n_1 \sin i = n_2 \sin r$

OR $\sin r = \dfrac{1.44 \sin 30^0}{1} = 0.72$

Now. $\angle\delta = \angle i = \sin^{-1}(0.72) - 30^0$

$\therefore \theta = 2(\angle\delta) = 2\{\sin^{-1}(0.72) - 30^0\}$

15. $[b] = \left[L^2\right]$

$$[a] = \frac{\left[L^2\right]}{[P][T]} = \frac{L^2}{\left[ML^2T^{-3}\right][T]} = \left[M^{-1}T^2\right]$$

$$[a \times b] = \left[M^{-1}T^2\right]\left[L^2\right] = \left[M^{-1}L^2T^2\right]$$

16. $\dfrac{47.5 - T}{42.5 - T} = \dfrac{8}{5}$

17. Radius of circular orbit $R = \dfrac{mv}{qB} = \dfrac{\sqrt{2mKE}}{qB} = \dfrac{\sqrt{2mT}}{qB}$

If T becomes double & 'B'' becomes tripled then radius becomes $\sqrt{\dfrac{2}{9}}R$

18. $V = \dfrac{ka}{r} = \dfrac{9 \times 10^9 \times 5 \times 10^{-9}}{15 \times 10^{-2}} = 300V$

19. $U = 10 - 10\cos 6x$

$\therefore F = -\dfrac{dU}{dx} = -60\sin 6x \approx -360x$

(for small x) $\therefore a = -\dfrac{360}{1}x \Rightarrow \omega^2 = 360$

$\therefore T = \dfrac{2\pi}{\omega} = \dfrac{1}{3}S$

$\therefore \ (A)$

20. $\lambda \propto \sqrt{\dfrac{1}{MT}}$

21. $6 \times 10^{-10}T$

22. Here centripetal force is provided by friction so

$\dfrac{mv^2}{r} \le \mu mg \Rightarrow v_{max} = \sqrt{\mu rg} = \sqrt{120} \approx 11ms^{-1}$

23. $E = W + K.E$

24. Energy required to remove first electron is $24.6eV$. After removing first electrons from this atom, it will become He^+

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$E_1 = -(13.6)(2)^2$ ($\left(as\ E \propto Z^2\ and\ Z = 2\right)$

$= -54.4eV$

$\therefore$ Energy required to remove this second electrons will be $54.4eV$.

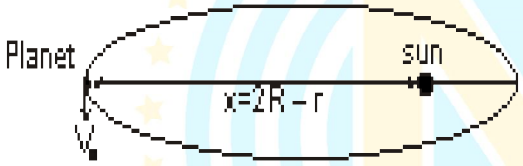
$\therefore$ Total energy required to remove both electrons

$= 24.6 + 54.4$

$= 79eV$

25. Area covered by line joining planet and sun in time dt is $ds = \frac{1}{2}x^2 d\theta$;

Area; velocity $= \frac{ds}{dt} = \frac{1}{2}x^2\frac{d\theta}{dt} = \frac{1}{2}x^2\omega$ where x = distance between planet and sun and □ = angular speed of planet about sun. From Keplers second law Areal velocity of planet is constant.



At farthest position $A = \frac{ds}{st} = \frac{1}{2}(2R-r)^2\theta = \frac{1}{2}\left[(2R-r)\omega\right] = \frac{1}{2}(2R-r)V_B$ or

$V_B = \frac{2A}{2R-r}$ (least speed). (Using values) $V_B = 4\,km/s$

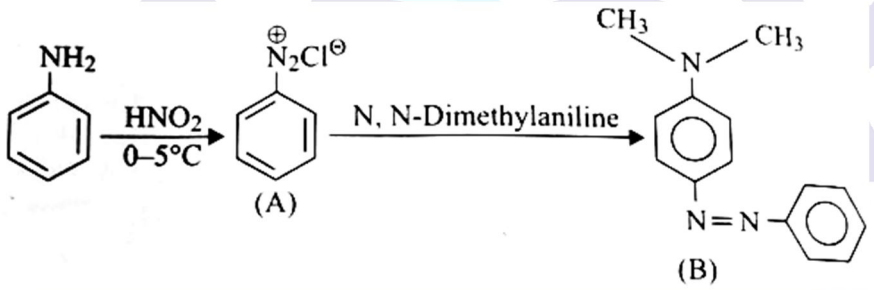
CHEMISTRY

26. Reducing power $\propto$ Tendency to undergo oxidation $\propto \frac{1}{reduction\ potential\ values}$

S0, reducing power decreases as $Ca > Mg > Zn > Ni$

27. Rate $= K\left[(C_6H_5)_3Cl\right]^1\left[OH^-\right]^0$

28.



29. For single electron species the energy depends upon principal quantum number 'n' only. So, statement II is false.
Statement I is correct definition of degenerate orbitals.

30. $\underset{sp^2}{CH_2} = \underset{sp^2}{CH} - \underset{sp^2}{CH} = \underset{sp^2}{CH_2}$

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31.

$$2H_3PO_3 + 4NaOH \rightarrow 2Na_2HPO_3 + 4H_2O$$

(A) (B)

$$\frac{M_AV_A}{2} = \frac{M_BV_B}{4} \Rightarrow \frac{50\times1}{2} = \frac{1\times x}{2} \Rightarrow x = 100\,mL$$

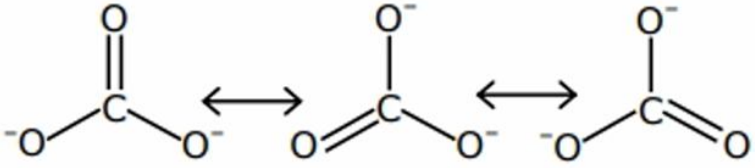
H_3PO_2 is monobasic acid so complete equation is

$$H_3PO_2 + NaOH \rightarrow NaH_2PO_2 + 2H_2O$$

(A) (B)

$$\frac{M_AV_A}{1} = \frac{M_BV_B}{1} \Rightarrow \frac{100\times2}{1} = \frac{1\times x}{1} \Rightarrow x = 200mL$$

32. Among SO_3, O_2, SO_2 AND CO_3^{2-} only O_2 and CO_3^{2-} has only one $\pi-$bond



33. Conceptual practical in organic chemistry group radicals & reageats

34. K_p change with T

35. $Z = 25 \Rightarrow Mn$

$E.C. = [Ar]^{18}\,4s^2 3d^5$

$Mn^{2+}, E.C = [Ar]^{18}\,3d^5$

$\Rightarrow$ No. of unpaired electrons = 5

$\Rightarrow \mu = \sqrt{n(n+2)}\,B.M.$

$\mu = \sqrt{5(5+2)}\,B.M.$

$\mu = \sqrt{35}\,B.M = 5.92\,B.M.$

36. (C) Molecular formula = (Emerical formula)n

$$n = \frac{Molecular\ weight}{Emperical\ \ formula\ wt.} = \frac{180}{30} = 6$$

$$= (CH_2O)_6 = C_6H_{12}O_6$$

37. $\lambda_{red} > \lambda_{green} > \lambda_{blue}$

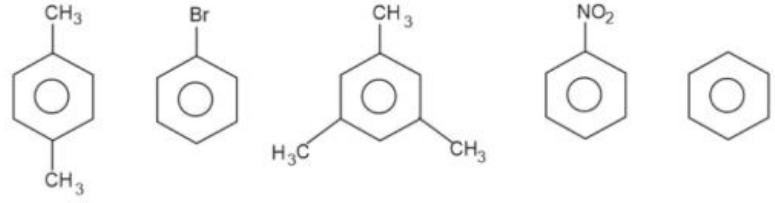
Order of λ_{abs} $L_3 > L_1 > L_2$

So order of ligand strength will be $L_2 > L_1 > L_3 \left(as\,\Delta_0 \propto \frac{1}{\lambda_{abs}} \propto E \right)$

38. Conceptual

39. Reactivity towards nitration $\propto$ Electron density on benzene ring $\propto +I, +R$ group

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A	B	C	D	E
P-xylene	Bromobenzene	mesitylene	nitrobenzene	Benzene

$-NO_2$ is deactivatiing
$-Br$ is deactivating
$-CH_3$ is activating group.

$D < B < E < A < C$ is the increasing order of reactivity towards nitrogen.

40. Conceptual

41. $C(graphite)+O_2(g)\rightarrow CO_2(g); \quad \Delta H_2^0 = -y\,kJ/mol$

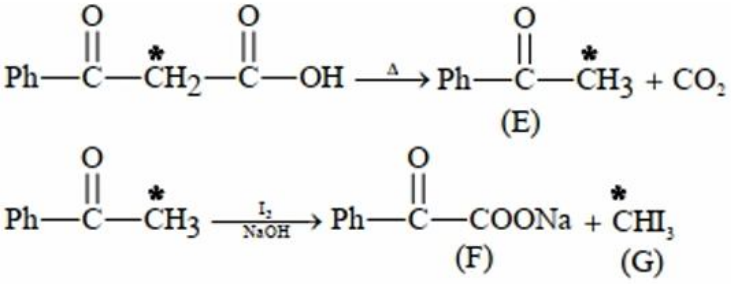
$\left[2CO_2(g)\rightarrow 2CO(g)+O_2(g)\right]\times\frac{1}{2};\ \Delta H_1 = -xkJ/mol\times\frac{1}{2}$

$C(graphite)+\frac{1}{2}O_2(g)\rightarrow CO(g)$

So, ΔH for required equation $=-y-\left(\frac{-x}{2}\right)=-y+\frac{x}{2}$

$=\frac{x-2y}{2}kJ/mol$

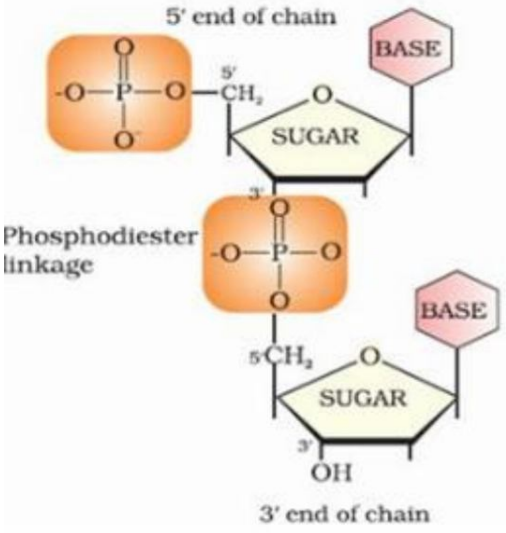
42.



43. Conceptual

44. Phosphodiester linkage

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45. $I.E_1$ of alkali metals is lowest and $I.E_2$ of alkali metals is highest because after losing one electron they acquire noble gas configuration so 'X' is Na.

46. Isotonic solution,

$\pi_{K_2SO_4} = \pi_{\text{Glucose}}$

$i \times 0.004 \times RT = 0.01 \times RT$

$i = 2.5$

For K_2SO_4 (for dissociation $i = 1 + (n-1)\alpha$)

$$DOD(\alpha) = \frac{i-1}{n-1} = \frac{2.5-1}{3-1} = 0.75$$

% dissociation = 75

47. Total $G.I = 2^2 = 4$

48. $$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right).$$

$k_2 = 10^{-3} s^{-1}, T_2 = 300K \quad k_1 = ?, T_1 = 200K$

$$\ln \frac{10^{-3}}{k_1} = \frac{11.488 \times 10^3}{8.314}\left(\frac{1}{200} - \frac{1}{300}\right)$$

$$\ln\left(\frac{10^{-3}}{k_1}\right) = \frac{11.5}{8.31} \times 10^3 \times \frac{100}{200 \times 300} = 2.303 = \ln 10$$

$$\Rightarrow \frac{10^{-3}}{k_1} = 10 \Rightarrow k_1 = 10^{-4} s^{-1}$$

$$\Rightarrow k_1 = 10 \times 10^{-5} s^{-1} \Rightarrow 10 \times 10^{-5}$$

49. $C = \frac{85.8}{12}$ $H = \frac{14.2}{1}$

$C_{7.15}H_{14.2} \Rightarrow CH_2$ $n \times EF_{mass} = MF_{mass}$

$14 \times n = 84$

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$n = 6$

$\therefore$, molecular formula $= C_6H_{12}$

50. % Bromine $= \dfrac{80 \times 0.12 \times 100}{188 \times 0.15}$

$= 34.04\%$

$= 34\%$

MATHEMATICS

51. Select Boys and Girls

GROUP A	GROUP B
1B 4G	3B
2B 3G	2B 1G
3B 2G	1B 2G
4B 1G	3G

52. Infinitely many solutions

$\Delta = 0$, $\Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Delta = \lambda\mu + 42\lambda - 4\mu + 107$

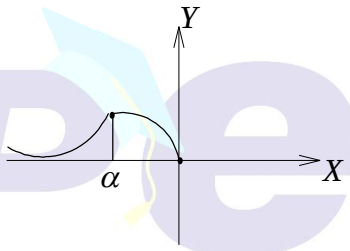
$\Delta_2 = \mu - 13$

We get λ and μ

53. Split the absolute value

based on domain 'x' Take 'K' as $\sqrt{x}$. Solve equation

54.



$y = e^x \ldots (1)$

$y = 1 - e^x \ldots (2)$

On solving we get $x = \alpha = -\ln 2$

$\therefore$ required area $= \int_{-\infty}^{-\ln 2} e^x dx + \int_{-\ln 2}^{0} \left(1 - e^x\right) dx = \ln 2$

55. Let eccentrically of conjugate hyperbola be e'

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$$\therefore \frac{1}{e^2}+\frac{1}{e'^2}=1 \Rightarrow \frac{1}{e'^2}=1-\frac{1}{e^2} \Rightarrow e'=\frac{e}{\sqrt{e^2-1}}$$

$$\therefore f(e)=\frac{e}{\sqrt{e^2-1}}$$

And $f(f(e))=\dfrac{f(e)}{\sqrt{f^2(e)-1}}=\dfrac{\dfrac{e}{\sqrt{e^2-1}}}{\sqrt{\dfrac{e^2}{e^2-1}-1}}=e$

$\therefore$ given integral

$$\int\left(\frac{e}{\sqrt{e^2-1}}+e\right)de=\left(\sqrt{e^2-1}+\frac{e^2}{2}\right)+C$$

Since $g(1)=\dfrac{1}{2} \Rightarrow C=0$

$$\therefore g(e)=\sqrt{e^2-1}+\frac{e^2}{2}$$

56. Using position of pt wrt lines (sides of triangle) we get range of α (common values)

57. $x^2 f'(x)=2xf(x)+3$

$$\frac{x^2 f'(x)-2xf(x)}{x^4}=\frac{3}{x^4}$$

$$\int d\left(\frac{f(x)}{x^2}\right)=\int\frac{3}{x^4}$$

$$f(x)=\frac{-1}{x}+cx^2$$

58. $20 = 6(1+\dfrac{1}{7}+\dfrac{1}{7^2}+\ldots.\infty)+\dfrac{\alpha}{7}\left(1+\dfrac{2}{7}+\dfrac{3}{7^2}+\ldots.\infty\right)$

$$20 = 7+\frac{\alpha}{7}\left(\frac{7}{6}+\frac{7}{36}\right)$$

$\alpha = 36$

(or)

$\dfrac{a}{1-r}+\dfrac{dr}{(1-r)^2}=14$; where $a=6, d=\alpha, r=\dfrac{1}{7}$

After simplification, we get $\alpha=36$

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59. $f(x)=\begin{cases}-x-1, & -2<x<-1\\ -x, & -1\le x<0\\ 1-x, & 0\le x<1\\ x, & 1\le x<2\\ x+1, & 2\le x<3\end{cases}$ which is discontinuous and non differentiable for 4 values of x they are $-1,0,1,2$

60. n (S) = 3^9 , n(E) = 3^6

61. Centroid = $\frac{\bar{p}+\bar{q}+\bar{r}}{3}$

G divides $\overline{OS}$ in the ratio $2:1\Rightarrow 3G=O+2S$

62. $\bar{b}=\bar{a}\times\left(\hat{i}-2\hat{k}\right)$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -1 & 1\\ 1 & 0 & -2\end{vmatrix}=2\hat{i}+5\hat{j}+\hat{k}$$

Find $\bar{c}$ and find projection

63. Graph

64. Given $f(x)=\frac{e^{2x}+1}{e^{2x}-1}$

$f(x_1)=f(x_2)\Rightarrow x_1=x_2$

Let $y=\frac{e^{2x}+1}{e^{2x}-1}\Rightarrow e^{2x}=\frac{y+1}{y-1}>0$

$\Rightarrow y<-1(or)\,y>1$

65. $20\{2a+39d\}=970$

$6\{2a+11d\}=39$

After simplification, $d=\frac{3}{2};a=-5$

66. General term of $(1+x)^n$

$T_{r+1}={}^nC_r\,x^r$

$2A=5B\Rightarrow 2\times{}^{(2n-1)}C_{29}=5\times{}^{(2n-1)}C_{11}$

$\frac{{}^{2n}C_{30}}{{}^{(2n-1)}C_{29}}=\frac{2n}{30};\frac{{}^{2n}C_{12}}{{}^{(2n-1)}C_{11}}=\frac{2n}{12}$

after simplification we get

n = 21

67. f (x) is strictly increasing Find the graph to check interval Find 'a'

Use wavy curve method for g (x) strictly decreasing we get 'b' , 'c'

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68. $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=1$

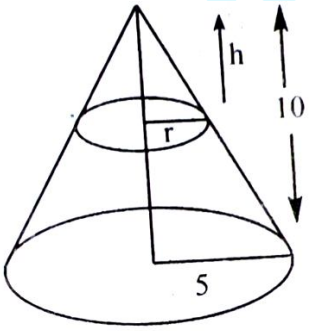
$$f(x)=\begin{vmatrix} a-t & 1 & b \\ a & 1+t & b \\ a & 1 & b-t \end{vmatrix} \text{ Where } t=\frac{\sin x}{x}$$

$R_2\to R_2-R_1; R_3\to R_3-R_1$

$\Rightarrow f(x)=-t^2(1+a+b-t)$

$=-(1+a+b)\left(\dfrac{\sin x}{x}\right)^2+\left(\dfrac{\sin x}{x}\right)^3$

69.



$\dfrac{dv}{dt}=3m^3/min$

$\dfrac{r}{h}=\dfrac{5}{10}=\dfrac{1}{2}$

$r=\dfrac{h}{2}$

$v=\dfrac{1}{3}\pi\times\dfrac{h^2}{4}\times h=\dfrac{\pi h^3}{12}$

$\dfrac{dv}{dt}=\dfrac{3\pi h^2}{12}\cdot\dfrac{dh}{dt}$

$\dfrac{dh}{dt}=\dfrac{4}{3\pi}$

70. $\cot^{-1}\left(\dfrac{\alpha^2+1-\alpha^2+\alpha\beta}{\alpha-\beta}\right)+\cot^{-1}\left(\dfrac{\beta^2+1-\beta^2+\beta^2}{\beta-\gamma}\right)+\cot^{-1}\left(\dfrac{\gamma^2+1-\gamma^2+\alpha}{\gamma-\alpha}\right)$

$\cot^{-1}\left(\dfrac{1+\alpha\beta}{\alpha-\beta}\right)+\cot^{-1}\left(\dfrac{1+\beta^2}{\beta-\gamma}\right)+\cot^{-1}\left(\dfrac{1+\alpha\gamma}{\gamma-\alpha}\right)$

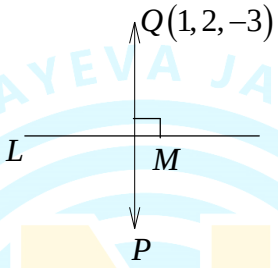
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$$= \cot^{-1}\left(\frac{1+\alpha\beta}{\alpha-\beta}\right)+\cot^{-1}\left(\frac{1+\beta\gamma}{\beta-\gamma}\right)+\cot^{-1}\left(-\left(\frac{1+\alpha\gamma}{\alpha-\gamma}\right)\right)$$

$$\cot^{-1}\alpha-\cot^{-1}\beta+\cot^{-1}\beta-\cot^{-1}\gamma+\pi-\left(\cot^{-1}\alpha-\cot^{-1}\gamma\right)$$

$$=\pi$$

71.



$$M=(2\lambda-1,-2\lambda+3,-\lambda)$$

Dr's of $\overline{QM}=2\lambda-2,-2\lambda+1,-\lambda+3$

Dr's of $L=2,-2,-1$

$\overline{QM}\perp L\Rightarrow a_1a_2+b_1b_2+c_1c_2=0$

$\Rightarrow \lambda=1$

Foot of $\perp r$, $M=(1,1-1)$

$P=2M-Q=(1,0,1)$

Since $R(-1,a,b)$ lies on $\frac{x+1}{2}=\frac{y-3}{-2}=\frac{z}{-1}$ then we get $a=3,b=0$

Area of $\Delta PQR=\frac{1}{2}\left|\overline{PQ}\times\overline{PR}\right|$

$$=\frac{1}{2}\left|10\bar{i}+8\bar{j}+4\bar{k}\right|=\sqrt{45}$$

72. $G.I=\int\frac{\left(2x^2+5x+9\right)}{\sqrt{x^2+x+1}}dx$

$$=2\int\sqrt{x^2+x+1}\,dx+\int\frac{3x+7}{\sqrt{x^2+x+1}}dx$$

$$=2.\int\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}\,dx+\frac{3}{2}.\int\frac{2x+1}{\sqrt{x^2+x+1}}dx+\frac{11}{2}\int\frac{1}{\sqrt{x^2+x+1}}dx$$

$$=2\left\{\frac{\left(x+\frac{1}{2}\right)}{2}.\sqrt{x^2+x+1}+\frac{3}{8}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|\right\}+3.\sqrt{x^2+x+1}$$

$$+\frac{11}{2}.\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+c$$

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On comparing $\alpha = \frac{7}{2}; \beta = \frac{25}{4}$

73. Divide the equation through by $\cos^2 y$

$\sec^2 y \frac{dy}{dx} + 2x.\tan y = x^3$

$\Rightarrow \frac{dz}{dx} + 2x.z = x^3$ where $z = \tan y$

$IF = e^{\int 2x dx} = e^{x^2}$

General solution is $z \times e^{x^2} = \int x^3 . e^{x^2} dx$

$= \frac{1}{2} fx^2 . e^{x^2} . 2x\, dx$

$\Rightarrow z.e^{x^2} = \frac{1}{2} . \int t\, e^t dt$

$\Rightarrow = e^{x^2} = \frac{1}{2} . e^{x^2} . \left(x^2 - 1\right) + c \Rightarrow \tan y = \frac{x^2 - 1}{2} + c.e^{-x^2}$

curve is $\tan y = \frac{x^2 - 1}{2} + \frac{3}{2} . e^{-x^2}$

74. f (1) + f (2) = f (3)

1	1	2
2	2	4
3	3	6
1	2	3 – 2 ways
1	3	4 – 2 ways
1	4	5 – 2 ways
1	5	6 – 2 ways
2	3	5 – 2 ways
2	4	6 – 2 ways
		Total = 15 ways

Number = (15) (6) = 90 functions

75. H : $\frac{x^2}{a^2} - \frac{y^2}{1} = 1$ E : $\frac{x^2}{4} + \frac{y^2}{3} = 1$

Latusrectum = $\frac{2}{a}$ Latusrectum = 3 a = $\frac{2}{3}$

$e_H = \sqrt{1 + \frac{9}{4}}$ $e_H = \sqrt{1 - \frac{3}{4}}$ $\Rightarrow e_E^2 = \frac{1}{4}$

$e_H^2 = \frac{13}{4}$ $12\left(e_E^2 + e_H^2\right)$ = 42

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